

30.2 A 22nm 0.26nW/Synapse Spike-Driven Spiking Neural Network Processing Unit Using Time-Step-First Dataflow and Sparsity-Adaptive In-Memory Computing

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 Contents

Recently, brain-inspired spiking neural networks (SNNs) have demonstrated tremendous improvement in energy efficiency (EE) and low power by exploiting highly sparse spikes and event-driven design [1– 2]. (top of Fig.30.2.1) With the same spike-based information carrier, the combination of an SNN and dynamic vision sensor (DVS) [3] offers a promising solution for edge AI applications. Typically, the greater the number of synapses in an SNN can improve inference accuracy, but induce more significant power consumption and memory size. Therefore, deploying more synapses in an SNN chip while maintaining low power, memory size and high EE requires co-design of algorithm, architecture and circuits. At the algorithm level, SNN training based on conversion from an artificial NN (ANN) [1] achieves the best accuracy in static image classification, but at the cost of worse EE and higher power consumption, as it requires excessive time steps and energy arising from ANN-SNN conversion. At the architecture level, the multiple-time-step forward computation in SNNs introduces redundant data movement and storage, including input-output spikes, weights, partial sums and membrane potentials, restricting the network scale that can be accommodated by limited on-chip memory, i.e., network density (ND) [4– 6]. At the circuit level, in-memory computing (IMC) is attractive to minimize data movement. However, traditional frame-based IMC circuits are incompatible with the highly sparse and fine-grained spike-driven nature of SNNs. Recent advancements in neuromorphic IMC [7] employ integrated-fire (IF) converters, which are globally triggered by power-hungry clocks even if there is no spike, resulting in high power per synapse, i.e., power density (PD).



This paper presents (Fig. 30.2.1) a comprehensively data-stationary (CDS) and spike-driven SNN chip featuring four sparsity-aware IMC macros to improve EE and reduce PD with high ND through architecture and circuit co-innovations. By leveraging the temporal-spatial information inherent in spikes, residue convolutional SNNs are applied with enhanced network scalability, achieving leading accuracy on both static and dynamic datasets with dramatically fewer weights [9]. 1) The proposed CDS architecture maximizes temporal-spatial data locality by fusing IMC with time-step-first (TSF) dataflow. Considering the limited on-chip memory, TSF dataflow keeps membranes spatially stationary and reuses local weights cached in the IMC macro during all time steps, achieving high ND. 2) We design a sparsity-adaptive IMC macro with self-triggering converters, which enables dynamic adaptation in power consumption and latency based on input spike sparsity, while liberating IMC from reliance on a global power-hungry clock. 3) A fine-grained spike-driven clock management system utilizes multi-level clock-gating and an asynchronous protocol, reducing energy consumption of “dark silicon” by $12.6\times$. 4) The four IMC macros can be flexibly combined according to different workloads for various scenarios. Notably, they can be configured to expand the maximum length of vector-matrix multiplication, thereby avoiding the costly multiple time-step movement of weights.

Figure 30.2.2 (top) shows the overall architecture, mainly comprising a TSF-dataflow controller, 64KB feature memory to store address-event-representation (AER) spikes, spike-fetching module (SFM), spike-reusing module (SRM), computation engine (CE), non-linear module (NLM) and spike-writing module (SWM). According to TSF dataflow, the TSF-dataflow controller functions as the top controller responsible for generating triggering signals for sub-controllers in other modules. Meanwhile, the SFM is responsible for sending required AERs to CE. A zero detector is employed to skip features where no spikes appear in the feature map (X_{in}, Y_{in}, T_i) , reducing data movement from input spikes. The SNN employs 8KB register files to cache reused spikes. The CE consists of 4 IMC macros, with outputs flexibly combined using an adder bridge. In each macro, a start-finish detector asynchronously triggers MAC operations, which is sparsity-adaptive realized by the IMC array and self-triggering converters. The calibration-batch-normalization (BN) module corrects and normalizes the raw MAC into MAC_{BN} . NLM performs leak IF (LIF) or residue operations. In NLM, each fired spike is encoded into AER to drive SWM. Being spike-driven, the SWM executes max-pooling or writes it back to SRAM directly. To minimize energy consumption on “dark silicon”, a fine-grained clock management system is proposed with a 4-level controller (Fig. 30.2.2 bottom). Combined with the spike-driven design, the whole system only enables one spike-wise grain, while all other grains are silenced.

Figure 30.2.3 shows the dataflow to realize CDS. During input fetching (Fig. 30.2.3 top), the input range is determined by the current output indexes. Then, all the pixels within the input range are shifted sequentially. The (X_{in}, Y_{in}) with no AER is skipped directly. Moreover, if the AERs are reused due to overlapping input range during output shifting, they are sent to both CE and SRM. The spike router in each IMC determines whether to accept current AERs based on the predefined (X_{in}, Y_{in}) and channel range (Fig. 30.2.3 middle-right). If a convolution involves excessive channels, the IMC macros are recombined to support more input channels, thereby mitigating the multiple-time-step data movements of partial sums and weights under limited computational resources. Each received AER by the router drives the IMC macro to load spike and read weights locally from subarray (Fig. 30.2.3 bottom-right). In TSF dataflow (Fig. 30.2.3 middle-left), the system continues shifting to the next output index until all required operations over all time steps have been completed. In each time step, the spike-driven local-weight-reading mechanism of the IMC engine significantly reduces the energy consumption of weight movement.

Figure 30.2.4 shows the implementation of the IMC array. It mainly consists of 8 KIMC cells, 9b-SRAM subarrays, and 128 self-triggering converters. The locally read 1b weight and 1b input (sin) determine whether C_0 is connected to the analog summation line (AS). Thus, the total capacitor (C_{mac}) in AS is determined by the current MAC results. During conversion (Fig. 30.2.4 bottom-right), each converter resets the AS to V_{DD} by enabling the start signal (from start detector). Then, the pulse generator is triggered to realize comparison between V_{AS} and V_{ref} . If $V_{AS} > V_{ref}$, the pulse generator is triggered again for the next comparison. Before comparison, V_{AS} is reset and the charge on C_{mac} is shared with C_{load} . If $V_{ref} > V_{AS}$, the converter generates a finish flag to the finish detector. The # of enabled conversions is referred to as the digital output. When all converters finish, the finish detector sends the final finish signal. The waveform of the IMC array (Fig. 30.2.4 bottom-right) shows how the adaptive energy and latency change under different input sparsity levels. Moreover, the self-triggering converter significantly reduces power consumed while waiting for events, due to its clock-free design. The top right of Fig. 30.2.4 shows the calibration-BN algorithm and its circuit implementation. We obtain the $gain_{mid}$ and $offset_{real}$ in each column by using the

two-point correction method. Then, $gain_{mid}$ is fused with the trained gain bn and the weights of binary bits to generate $gain_{real}$

[Figure 30.2.5](#) shows the measurement results of our SNN chip fabricated in 22nm CMOS. The IMC macro achieves high linearity and consistency with two-point calibration, which reduces RMS error from 15.05 to 4.8 (from 1.30 to 0.73 for sparse inputs), improving the classification accuracy on DVS-Gesture datasets from 89% to 94% ([Fig. 30.2.5](#) middle). Moreover, the energy consumption of the IMC is almost linear to non-zero density demonstrating sparsity-adapative characteristics suitable for SNNs. Thanks to fine-grained and spike-driven clock management, $> 95\%$ power from clock tree is saved at 0.8V when system is static. During inference, the system saves running power by $12.6\times$. Compared to state-of-the-art (SOTA) work, our residue SNN with measured linearity and quantization exhibits higher classification accuracy than [\[2\]](#) in the MNIST and DVS-Gesture datasets. Our measured EE-voltage characteristic during inference is shown in [Fig. 30.2.5](#). The peak EE is 3.78pJ/SOP at 0.55V and 51MHz.

[Figure 30.2.6](#) compares our chip with SOTA. We normalize the supported # of synapses with memory size to represent the ability to deploy NNs in limited memory space, i.e., ND. The running power is normalized with the supported # of synapses to eliminate the workload dependence of power consumption, i.e., PD. Due to the weight and membrane reusing in TSF dataflow, compared with SOTA, our chip improves ND (18.18 synapses/B) by $6\times$ and improves PD (0.26nW/synapse) by $2.3\times$. Furthermore, our chip achieves superior comprehensive performance considering both throughput and PD ([Fig. 30.2.6](#) top-right). More synapses will contribute larger static power consumption due many silent synapses in SNNs, worsening the EE. This chip achieves excellent EE (3.78pJ/SOP) while supporting more synapses (2M). In contrast, although [\[2\]](#) also has outstanding EE, it has fewer synapses. [Figure 30.2.7](#) shows the micrograph of the chip, summary table, frequency-voltage characteristic, and test platform. Our chip consumes 0.524mW at 51MHz/0.55V and 6.2mW at 280MHz/0.9V.



Figure 30.2.1:
The background of SNNs (top). The challenges and innovation of proposed designs.



Figure 30.2.2:
The overall architecture and multi-level clock management.



Figure 30.2.3:
IMC with time-step-first dataflow strategy (bottom-left). Sparsity utilization and data reuse (top-left). Flexible combination of IMC macros (middle-right). Spike-driven loading of weight and input over multiple time steps (bottom-right).



Figure 30.2.4:
Detailed circuit implementation of sparsity-adaptive and clock-free IMC macro (top-left). Diagram of calibration-BN fusion design (top-right). Waveform of the IMC with adaptive energy and latency (bottom-right).



Figure 30.2.5: Measurement and evaluation results of sparsity-adaptive IMC circuits (top-left), power measurement (top and middle-right), and the energy efficiency measurement (bottom-right). The applied network comparison and evaluation (bottom-left).



Figure 30.2.6: Comparison table and improvement.



Figure 30.2.7: Chip micrograph (top-left). Summary table (top-right). Voltage-power characteristics (bottom-left). Test platform (bottom right).

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